Tahmidul Bin Ferdous

Department of Computer Science and Engineering, International Islamic University Chittagong, Bangladesh

Chittagong, Bangladesh | tahmidul.ferdous.edu@gmail.com | +880-1630620988 | tahmidulferdous.github.io

GitHub: tahmidulferdous | LinkedIn: tahmidul-kashfi | Google Scholar | ORCID: 0009-0001-1459-4124

Research Interests

Primary: Technical AI Safety, Alignment, Evaluation Validity, Benchmark Integrity & Data Contamination Auditing.
Secondary: Generative Modeling (Flow Matching, Diffusion, GANs), Medical Computer Vision, Distributed Training (Multi-GPU DDP, TPU v5e/JAX).

Education

International Islamic University Chittagong (IIUC) *Chittagong, Bangladesh*
Bachelor of Science in Computer Science and Engineering *2022 – 2026*

- **Undergraduate Thesis:** Detecting Data Leakage in Lung Cancer CT Classification, across CNN, EfficientNet, and Vision Transformer models.
- Post-submission re-analysis corrected the thesis's central claim: the IQ-OTH/NCCD release carries no patient identifier, so the comparison is an **ordering split**, not a patient-level one, and the measured cost is **7.5 accuracy points** across three seeds rather than the 12.46 originally reported. Full code, run logs and figures: lung-ct-data-leakage-study.

Publications

- *Manuscripts under preparation.*

Technical Skills

- **Programming Languages:** Python (Primary, 250+ logged experiments), C++, SQL, Bash/Shell, LaTeX.
- **Deep Learning Frameworks:** PyTorch (Advanced), JAX/Flax, torchvision, timm, TensorFlow/Keras, Hugging Face.
- **Computer Vision & ML Libraries:** OpenCV, MediaPipe, Ultralytics YOLOv11, scikit-learn, Albumentations, NumPy, Pandas, Matplotlib.
- **Systems & Infrastructure:** Distributed Data Parallel (`torchrun`, multi-GPU DDP), Google Cloud TPU v5e (8-device JAX), Mixed Precision (AMP), Weights & Biases (250 tracked runs), Git, Docker, Linux/Ubuntu.
- **Models Built from Scratch:** Autoregressive Decoder Transformer (GPT-1), Rectified Flow / Flow Matching with ODE solvers, Conditional DDPM, WGAN-GP (gradient penalty & critic loop), DCGAN, Mamba-Attention hybrid (DSMaT).

Research & Professional Experience

EliteLab *Remote*
Research Intern *2026 – Present*

Outlier AI *Remote*
AI Data Quality & Alignment Specialist *November 2024 – February 2025*

- Audited and curated large-scale human preference and reasoning datasets utilized for Reinforcement Learning from Human Feedback (RLHF) and alignment.
- Refined multi-turn annotation guidelines, directly mitigating prompt ambiguity and improving inter-annotator reliability across reasoning benchmarks.

Freelance Contract (Client in USA) *Remote*
Computer Vision Engineer *November 2024*

- Designed and deployed an automated computer vision packet-counting system for inventory management within a 15-day timeline, achieving real-time execution (>30 FPS) under CPU constraints.

Research & Technical Projects

Conditional Rectified Flow for CT Synthesis on Google Cloud TPU v5e *Feb 2026 – Mar 2026*

PyTorch, JAX/Flax, Google Cloud TPU v5e, Weights & Biases (18 runs, 32.1 GPU-hours) | *Code*

- Implemented class-conditional Rectified Flow (flow matching) from original mathematical literature: linear interpolant trajectory, velocity-prediction objective, logit-normal timestep sampling, and Euler/midpoint ODE numerical solvers.
- Built multi-GPU DistributedDataParallel (DDP) PyTorch training infrastructure, then **ported the complete architecture to JAX/Flax for 8-device Google TPU v5e**, achieving **576 images/sec** (2.50 s/epoch vs. 15.5 s/epoch on 2× Tesla T4)—a ∼**6**× **throughput speedup**.

BanglaPoemGPT — 10M-Parameter Autoregressive Transformer from Scratch *March 2025*

Python, PyTorch, Kaggle P100 (4 runs, 2.0 GPU-hours) | *Code*

- Engineered a GPT-1 architecture from first mathematical principles without high-level library wrappers: multi-head causal self-attention, learned positional embeddings, custom tokenizer, and autoregressive generation loop.
- Trained on a 3M-token Bengali poetic corpus, achieving validation cross-entropy loss of **1.693** with coherent generative sampling.

Conditional DDPM for Minority-Class CT Balancing *February 2026*

PyTorch, DistributedDataParallel, 2× Tesla T4 (5 runs, 4.2 GPU-hours) | *Code*

- Constructed a class-conditional Denoising Diffusion Probabilistic Model (DDPM) featuring a conditional U-Net backbone, cosine noise schedule, and multi-GPU DDP orchestration, reaching a training loss of **0.0238** over 3,000 epochs.

Edge Vision: YOLOv11 Campus Security & Facial Biometric Systems *Oct 2024 – Nov 2024*

YOLOv11, OpenCV, FaceNet, PyTorch | *Code*

- Curated, annotated, and trained custom YOLOv11-s/m models for campus ID compliance monitoring, reaching **mAP50 = 0.82**.
- Implemented edge facial recognition utilizing SSD face detectors and FaceNet 128-d embeddings with cosine similarity matching, calibrating decision thresholds against false-accept/false-reject trade-offs.

All experiments above are published with their code, per-run metrics and figures: github.com/tahmidulferdous/ml-research-log — 250 tracked training runs across 14 projects.

Awards, Certifications & Achievements

- **DeepLearning.AI Specializations (Coursera):** Neural Networks & Deep Learning, Improving Deep Neural Networks (Tuning, Regularization, Optimization), Convolutional Neural Networks, Sequence Models.

Academic References

Md. Khaliluzzaman
Associate Professor & Undergraduate Thesis Supervisor
Department of Computer Science and Engineering
International Islamic University Chittagong (IIUC)
Chittagong, Bangladesh
Email: khaliluzzaman@iiuc.ac.bd

Dr. Abu Nowshed Chy
Assistant Professor
Department of Computer Science & Engineering
University of Chittagong (CU)
Chittagong, Bangladesh
Email: nowshed@cu.ac.bd